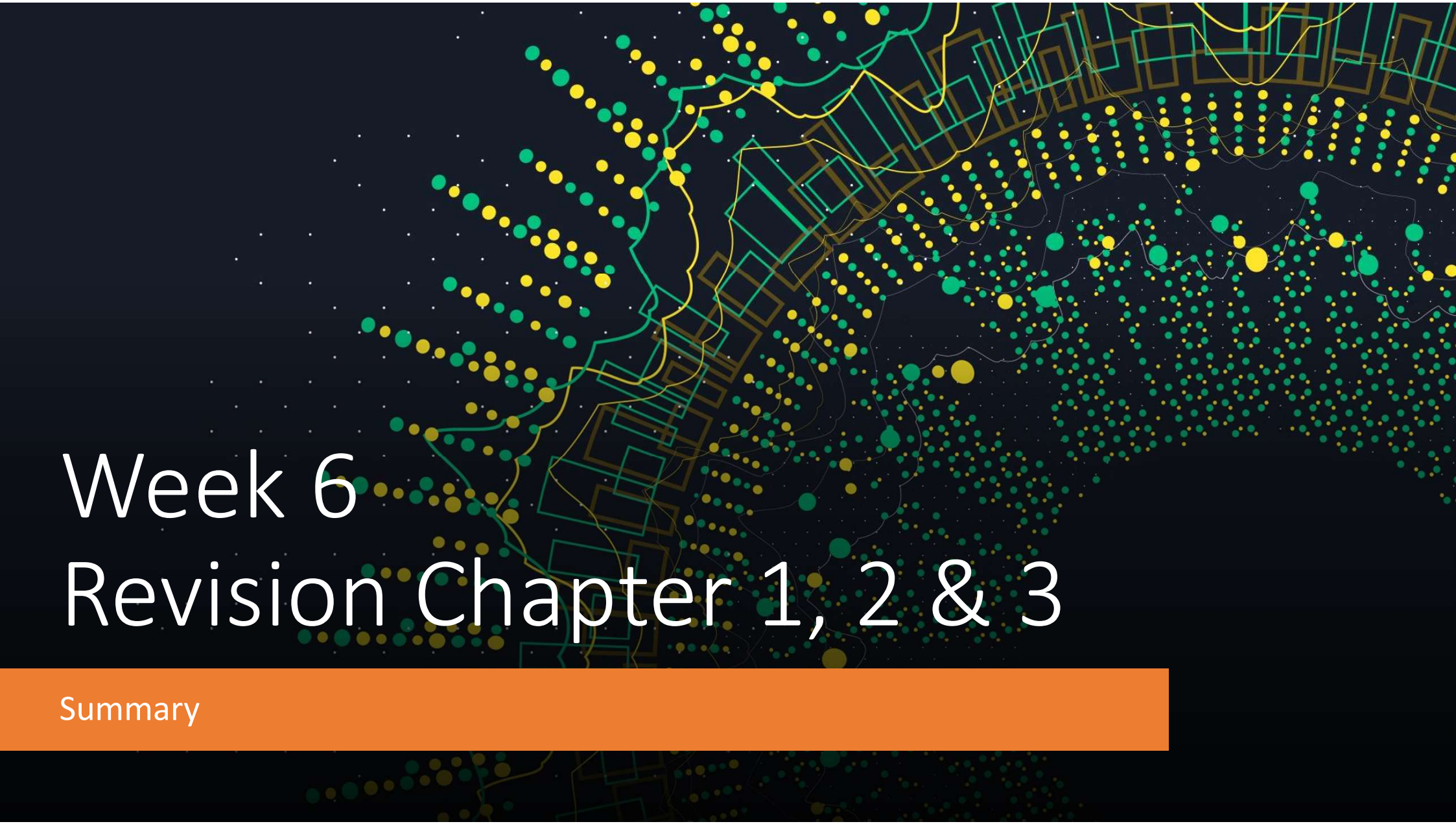


Week 6
Revision Chapter 1, 2 & 3

Summary

Format of Final Exam

(a) Definition	State the definition of … State the meaning of … 3 to 4 marks
(b) Theorem	Prove the theorem … (you are not required to remember the naming of the theorem) 6 to 7 marks
(c) Unseen	Prove or determine something (maybe) related to (b) Or use (b) to prove or determine other statements 6 to 7 marks
(d) True/false (subject to changes)	State true or false. Provide a proof if it is true or otherwise give a counter example. 3 to 4 marks

Chapter 1 Numbers

Definitions, theorems, examples

State the definitions of

1. A set which is bounded above/below.
2. An upper bound/lower bound.
3. Supremum/infimum.
4. Maximum/minimum.

Theorems

Archimedean property
$\mathbb{Q}$ is dense in $\mathbb{R}$
$\mathbb{R}\backslash\mathbb{Q}$ is dense in $\mathbb{R}$
Nested interval property
Supremum/infimum is unique
Suprema and Infima Analytically
Triangle Inequality and Reverse Triangle Inequality

Final exam style Q1

(a) State the definition of supremum.

(b) State and prove the Archimedean property.

(c) Guess the $\sup\{1 - 2/n : n \in \mathbb{N}\}$ if they exist. Then prove your guess is correct.

(d) Suppose set A is a nonempty subset of $\mathbb{N}$ that is bounded above. Prove or disprove that A has a supremum.

(a) Let $\mathbb{F}$ be an ordered field. Let $\varnothing \neq S \subseteq \mathbb{F}$ be a set that is bounded above. Then the supremum of S, if it exists, is the least upper bound of S.

(b) Archimedean property: Let $x, y \in \mathbb{R}$ such that $y > x > 0$. Then there exists $n \in \mathbb{N}$ such that $nx > y$.

Proof. Set $z = \frac{y}{x} \in \mathbb{R}$. Then note that $z > 1$. We prove that there exists $n \in \mathbb{N}$ such that $n > z$ for all $z > 1$ (by contradiction). Assume that there exists at least a $z > 1$ such that $n \leq z$ for all $n \in \mathbb{N}$. This means that $\mathbb{N}$ is bounded above. By the completeness axiom, since $\varnothing \neq \mathbb{N} \subseteq \mathbb{R}$, there exists a supremum $\alpha = \sup \mathbb{N}$.

We first show that $\alpha \notin \mathbb{N}$. Suppose not. Then $\alpha \in \mathbb{N}$ and this implies that $\alpha < \alpha + 1$. This leads to a contradiction as $\alpha \geq a \ \forall a \in \mathbb{N}$ (is an upper bound) and $\alpha + 1 \in \mathbb{N}$.

Now since $\alpha \notin \mathbb{N}$, it follows that $\alpha \in \mathbb{R} \setminus \mathbb{N}$. Then there exists an $m \in \mathbb{N}$ such that

$$\alpha - 1 < m < \alpha \,,$$

which implies that (from the left-hand side) $\alpha < m + 1 \in \mathbb{N}$, a contradiction, as α is an upper bound. The proof is completed.

(c) Note that $0 \in \left\{1 - \frac{2}{n} : n \in \mathbb{N}\right\}$ (when $n = 2$), so the set is non-empty. We guess $\sup\left\{1 - \frac{2}{n} : n \in \mathbb{N}\right\} = 1$. To prove it, we can use either one of the following:

Method 1: By Theorem 4.2(a), Chap 1.

1 is an upper bound since $1 > 1 - \frac{2}{n} \forall n \in \mathbb{N}$.

Suppose $\exists \varepsilon > 0$ such that $1 - \varepsilon \geq a \ \ \forall a \in \left\{1 - \frac{2}{n} : n \in \mathbb{N}\right\}$.

Then $1 - \varepsilon \geq 1 - \frac{2}{n} \Rightarrow \varepsilon \leq \frac{2}{n} \forall n \in \mathbb{N}$, contradicts to the Archimedean property.

Method 2: By Theorem 7.5, Chap 2.

1 is an upper bound since $1 > 1 - \frac{2}{n} \forall n \in \mathbb{N}$.

Construct $\left(a_n\right) = \left(1 - \frac{2}{n}\right)$. Then $\lim\left(a_n\right) = \lim\left(1 - \frac{2}{n}\right) = 1 = \sup\left\{1 - \frac{2}{n} : n \in \mathbb{N}\right\}$.

Method 3: By definition. 1 is an upper bound since $1 > 1 - \frac{2}{n} \forall n \in \mathbb{N}$.

Suppose $m = \sup\left\{1 - \frac{2}{n} : n \in \mathbb{N}\right\}$ and $m < 1$. Then $\exists \delta > 0$ such that $m = 1 - \delta$.

Now, m is an upper bound, which means $1 - \delta = m \geq 1 - \frac{2}{n} \Rightarrow \delta \leq \frac{2}{n} \forall n \in \mathbb{N}$, which contradicts to the Archimedean property.

(d) A has a supremum. Since $\varnothing \neq A \subseteq \mathbb{N}$ is bounded above, it follows that A is a finite set. Hence, $\max A$ exists and $\max A = \sup A$. So, the supremum exists.

Chapter 2 Sequences

Definitions, theorems, examples

Definitions

- Bounded sequence
- Unbounded sequence
- Convergent sequence
- Divergent sequence
- Monotone increasing sequence
- Monotone decreasing sequence
- Subsequence
- Cauchy sequence

The algebraic and order limit theorems

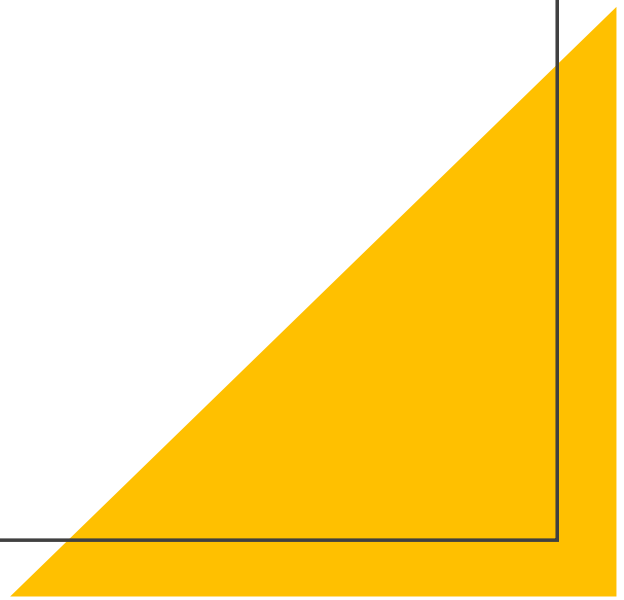
Theorem 6.1 (=)

Theorem 6.3 (<)

Theorem 6.4 (Squeeze)

Theorem that links sequence to sup/inf in sets

- Theorem 7.5: …



Theorem on convergence

Let (a_n) be a sequence.

Theorem 5.4. If (a_n) is convergent, then (a_n) is bounded.

Theorem 7.3. (i) If (a_n) is increasing and bounded above then (a_n) is convergent.

(ii) If (a_n) is decreasing and bounded below then (a_n) is convergent.

Theorem 8.3. (a_n) converges to L iff every subsequence of (a_n) converge to L.

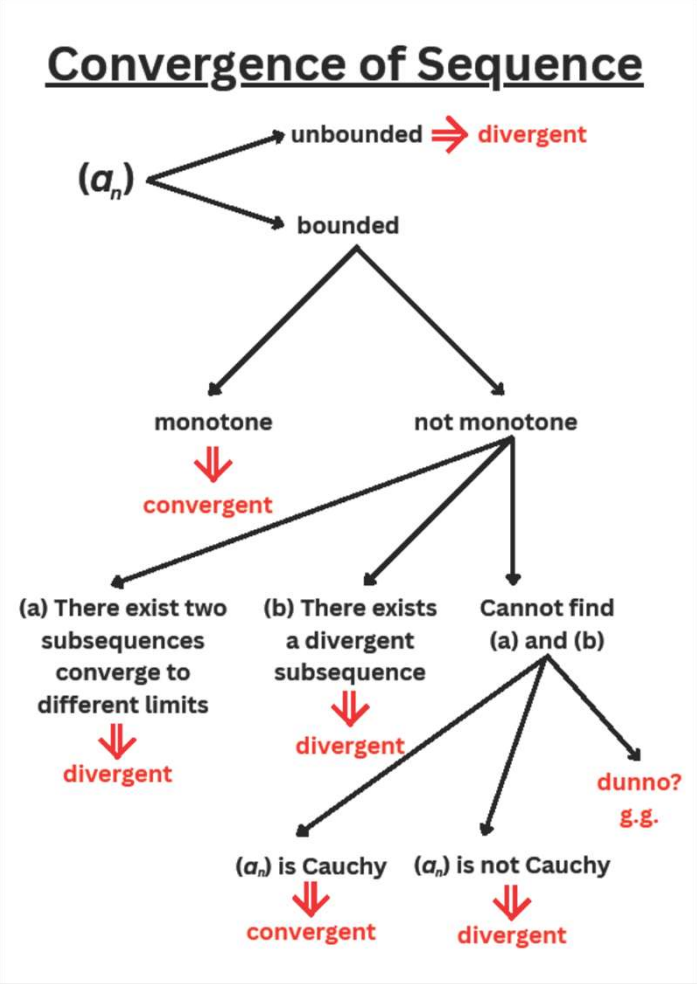
Corollary 8.4. If there are two subsequences of (a_n) converge to different limit, then (a_n) is divergent.

Theorem 8.6. Every sequence has a monotone subsequence.

Theorem 8.7. Every bounded sequence has a convergent subsequence.

Theorem 9.2. All Cauchy sequences are bounded.

Theorem 9.3. (a_n) is Cauchy iff (a_n) is convergent.



How to determine a sequence is conv/divergent?

Final exam style Q2

(a) State the definition of a convergent sequence that converges to $a \in \mathbb{R}$.

(b) State and prove the monotone convergence theorem.

(c) Let (a_n) be a sequence such that

$$a_{n+1} = \frac{1}{2}\left(a_n + \frac{5}{a_n}\right) \text{ with } a_1 = 3\,.$$

Prove that (a_n) is convergent and write $\lim(a_n)$.

(d) All bounded sequence is convergent. Provide a proof if it is true or otherwise give a counter example.

(a) A sequence $\left(a_n\right)$ is said to be convergent (and converges to $a \in \mathbb{R}$), if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $n > N$ implies that $\left|a_n - a\right| < \varepsilon$ for all $n \in \mathbb{N}$.

(b) Check the proof.

(c) Given $a_{n+1} = \frac{1}{2}\left(a_n + \frac{5}{a_n}\right), a_1 = 3$.

We show that $\left(a_n\right)$ is bounded below, i.e., $a_n > 0 \ \forall n \in \mathbb{N}$.

Note that $a_1 = 3 > 0$ is true. Assume $a_k > 0$ for some $k \in \mathbb{N}$.

Now, $a_{k+1} = \frac{1}{2}\left(a_k + \frac{5}{a_k}\right) > 0$. By induction, $\left(a_n\right)$ is bounded below.

We next show that $\left(a_n\right)$ is decreasing.

Note that $2a_{n+1} = a_n + \frac{5}{a_n} \Rightarrow 2\left(a_{n+1} - a_n\right) = -a_n + \frac{5}{a_n}$.

We need to prove that $-a_n + \frac{5}{a_n} < 0$, or $a_n^2 > 5$.

Note that $a_1^2 = 9 > 5$ is true. Assume $a_k^2 > 5$ for some $k \in \mathbb{N}$.

Now, $a_{k+1}^2 = \frac{1}{4}\left(a_k + \frac{5}{a_k}\right)^2 = \frac{1}{4}\left[\left(a_k - \frac{5}{a_k}\right)^2 + 20\right] \geq \frac{1}{4}(20) = 5$.

By induction, $a_n^2 > 5$ and so $\left(a_n\right)$ is decreasing.

By MCT, $\left(a_n\right)$ is convergent. Let $\lim\left(a_n\right)=a$. Note that $a \geq \sqrt{5}$.

Then $\lim\left(2a_{n+1}\right)=\lim\left(a_n+\frac{5}{a_n}\right) \Rightarrow a^2=5 \Rightarrow a=\sqrt{5}$.

(d) False. A counter example is $\left((-1)^n\right)$ which is bounded but diverge.

Chapter 3 Series

Definitions, theorems, examples

Definition

- Partial sum
- Series
- Convergent series
- Divergent series

Definitions & Series test

- Divergence test
- Geometry series
- Harmonic series
- *p* series
- Comparison test
- Limit comparison test
- Ratio test
- *n*-th root test
- Absolute convergence test
- Alternating series test
- Conditionally convergence series

Theorems and proofs

- Theorem 3.5. Series properties
- Theorem 4.1. Cauchy Criterion for Series
- Theorem 4.2. Divergence test
- Example 4.4. Harmonic series
- Theorem 4.7. Comparison test
- Theorem 4.5. Cauchy Condensation Test
- Theorem 4.6. p-series test
- Theorem 4.12. Absolute Convergence
- Theorem 4.16. Ratio Test
- Theorem 4.9. Alternating series test

Final exam style Q3

(a) State the definition of a divergent series.

(b) Prove that the harmonic series is divergent.

(c) Prove that the alternate harmonic series is convergent.

(d) Suppose $\sum a_k$ and $\sum b_k$ are convergent, then $\sum a_k b_k$ is convergent. Provide a proof if it is true or otherwise give a counter example.

(a) Check the definition, there are three cases.

(b) Check the proof.

(c) By using the alternating series test, $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is convergent, because

$\left(a_n\right) = \left(\frac{1}{n}\right)$ is decreasing since $a_n - a_{n+1} = \frac{1}{n} - \frac{1}{n+1} > 0 \ \forall n \in \mathbb{N}$,

and $\lim\left(a_n\right) = \lim\left(\frac{1}{n}\right) = 0$.

(d) False. $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ and $\sum_{n=1}^{\infty} \frac{(-1)^{n}}{\sqrt{n}}$ are convergence series but $-\sum_{n=1}^{\infty} \frac{1}{n}$ is not.